

Swastika Issar

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Research Experience

- 2021- Present Visiting Researcher
Max Planck Institute of Animal Behavior/ University of Konstanz
Funded by: The *Max Planck Society* & a *MediumMoney21 Grant from the Centre for the Advanced Study of Collective Behaviour*
Project: How does resource distribution impact social behaviour in the cichlid fish of Lake Tanganyika?
Project advisor: Dr. Alex Jordan
- 2016 – 2022 Graduate Student
Department of Zoology, University of Cambridge
Funded by: *Dr. Manmohan Singh Scholarship from St. John's College*
PhD Thesis: Ecological and molecular basis of differential resource use in populations of burying beetles *Nicrophorus vespilloides*
Supervisor: Prof. Rebecca Kilner
Advisors: Prof. Chris Jiggins, Prof. Andrea Manica
- 2016 Visiting Scholar
Department of Zoology, University of Cambridge
Funded by: *Sir Dorabji Tata Trust Travel Grant*
Project: How do burying beetles use chemical signals from microbes to locate reproductive and nutritional resources?
Supervisor: Prof. Rebecca Kilner
- 2013 – 2016 Project Assistant
National Centre for Biological Sciences, Bangalore
Project(s):
- Genetic and phenotypic variation in natural populations of the red flour beetle across the Indian subcontinent
- Cannibalistic behaviour in natural populations of flour beetles
- Effect of prior experience on resource preference in red flour beetle larvae
Principal investigator: Dr. Deepa Agashe
- 2013 Master's Project Intern
National Centre for Biological Sciences, Bangalore
- Studied the peptide toxin Glacontryphan M in the Indian subspecies of the Great Orange Tip butterfly
- Compared bioactive peptides isolated and characterized from Indian butterfly species
Project advisor: Dr. Deepa Agashe, Dr. K. S. Krishnan & Dr. Krushnamegh Kunte
- 2012 Summer Scholar
National Centre for Antarctic & Ocean Research, Goa
Project: Community composition and thermal adaptations of microbial species in an arctic fjord (Kongsfjorden)
Project advisor: Dr. K.P. Krishnan

Education

- 2016 – Ph.D in Zoology
2022 Department of Zoology, University of Cambridge, United Kingdom
- Collaborative work at Université Paris funded by a Santander International Mobility Grant and the Graduate Scholar's Research Scheme at St. John's College
- 2011 – M.Sc Environmental Studies
2013 TERI University, New Delhi, India
- Cumulative GPA 7.8
- Independent study project on microbe-plant interactions at the School of Life Sciences, Jawaharlal Nehru University, New Delhi, India
- 2009 – B.Sc (Honours) Biological Sciences
2011 University of Delhi (Sri Venkateswara College), New Delhi, India
- Secured First Division both cumulatively and during the individual years of study.
- 2008 – B.Sc (Program) Life Sciences
2009 University of Delhi (Sri Venkateswara College), New Delhi, India
- Selected for migration to the Honours Program in Biological Sciences on receiving First Division in First Year examinations

Publications

Issar S (2021). Ecological and molecular basis of differential resource use in populations of burying beetles *Nicrophorus vespilloides* (Doctoral thesis)

Khan I, Prakash A, **Issar S**, Umarani M, Sasidharan R, Masagalli JN, Lama P, Venkatesan R and Agashe D (2018). Female chemical warfare drives fitness effects of group sex ratio. *American Naturalist* 191(3):306–317

Ravi Kumar V, **Issar S** and Agashe D (2018). The induction of larval resource preference in heterogeneous habitats. *Ecological Entomology* 43: 719–730

Upcoming publications from long-term field studies and experimental evolution work during PhD

Jarrett BJM, Mashoodh R, **Issar S**, Pascoal S, Rebar D, Sun S-J, Schrader M and Kilner RM (2022). Multilevel selection leads to divergent coadaptation of care-giving parents during pre-hatching parental care. Preprint. Biorxiv

Issar S, d'Ettore P, Sheehan M and Kilner R. (In prep) The adaptations of burying beetles to seasonal environments

Issar S, Park H-W and Kilner R. (In prep) Evolutionary consequences of adaptation to resource-poor environments

Awards and grants

- 2021 MediumMoney21 Grant
2020 Leche Trust Grant
2019 Cambridge Philosophical Society Research Studentship
2019 Santander International Mobility Grant
2016 - 2019 Dr Manmohan Singh Scholarship
2016 - 2019 Trinity-Henry Barlow Honorary Scholarship
2016 Hitchcock Fund Award
2016 Sir Dorabji Tata Trust Travel Grant

Teaching roles

Feb 2021	Supervision and examination assistant for evolution and behaviour undergraduate practicals at the Department of Zoology, University of Cambridge
Oct - Nov 2019	STIMULUS program Biology Teaching Assistant at Long Road Sixth Form College, Cambridge
May 2019	Assistant supervisor for undergraduate animal physiology practicals at the Department of Zoology, University of Cambridge

Conference presentations

2020	Virtual poster presentation at Animal Behaviour Live Poster: How does resource specialisation evolve?
2019	Invited speaker at St John's College Graduate Research Day (Cambridge, United Kingdom) Talk: <i>Isolation between populations evolving in different social environments- predictions and experimental outcomes</i>
2018	Nominated speaker at Zoology Graduate Symposium (Cambridge, United Kingdom) Talk: <i>Social epistasis and reproductive isolation in burying beetle populations</i>
2017	Awarded best lightning talk at BES Ecology Across Borders conference (Ghent, Belgium) Talk: <i>Resource use and reproductive isolation in burying beetles</i>
2016	Poster at the India-Behaviour, Ecology & Evolution Conference (Uttarakhand, India) Poster: <i>Resource choice in the red flour beetle larvae: How experience affects foraging behaviour</i>
2014	Designed and presented an experimental evolution protocol at the TULIP LabEx Summer School in Integrative Biology and Ecology (Toulouse, France) Talk: <i>Understanding mutualistic interactions and how they change with environmental alterations using the ant- aphid system</i>
2014	Student research talk at the School and Discussion Meeting on Population Genetics and Evolution organized by ICTS, TIFR (Bangalore, India) Talk: <i>Fitness consequences of larval resource choice in the flour beetles</i>

Leadership and outreach roles

2020-Present	STEM mentor at The Bhagini Project, a not-for-profit women's collective that connects Indian women to mentors in their field of interest to act as a support system in their application process to graduate programs
2021	First and second editor; Issues 51 & 52 of BlueSci, the science communication magazine at the University of Cambridge
2018 - 2019	Podcast host and news editor at BlueSci
2017	Burying beetle 'expert' at the Just Bugs! event of the Cambridge Science Festival
2017 - 2018	Graduate representative at the Department of Zoology
2016 - 2017	Cambridge Science Society college and department representative
2016	Volunteer at Soapbox Science, a public outreach platform for promoting women scientists and their science

Academic referees

Prof. Rebecca Kilner

Professor of Evolutionary Biology

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University of Cambridge
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Cambridge CB2 3EJ
United Kingdom

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Dr. Deepa Agashe

Assistant Professor

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Tata Institute of Fundamental Research
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Bangalore 560065
Karnataka, India

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